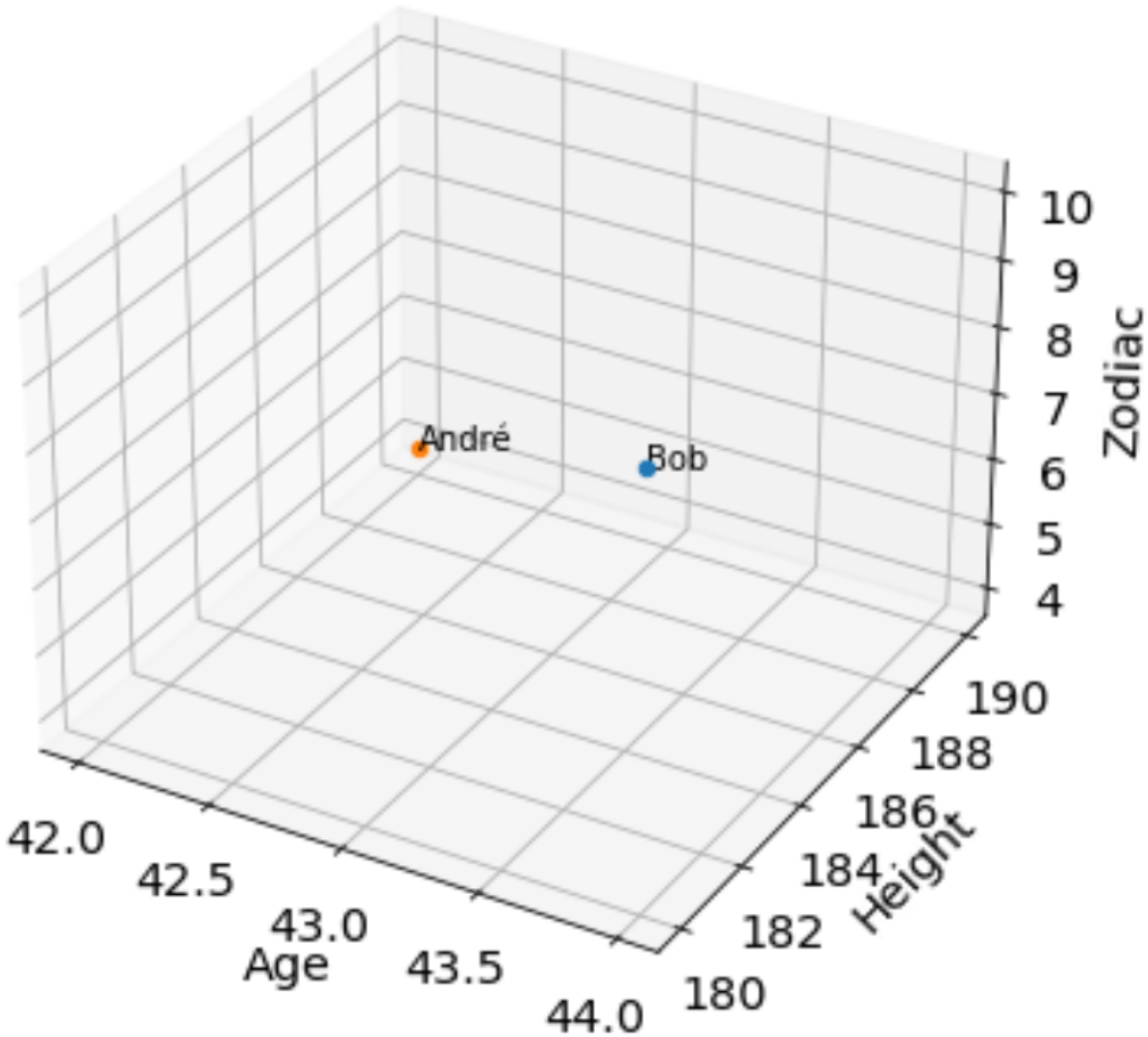
Lab 2: Machine Learning Fundamentals

In this lab you will practice some of the fundamentals of machine learning: vectors. You will also continue to practice programming in python, which is another kind of "fun". Complete all parts marked with a "*" at least. You can do more parts, of course, because in the end that will help you on your project!

1.1*

Build two vectors that numerically describe you and your lab partner. Make the two vectors have the following dimensions in this order: age (years), height (cm), zodiac sign (coded 0-11) with 0 being Aquarius and 11 being Capricorn. Plot them in a three dimensional plot. Make sure you label the axes! This is what mine looks like with my lab partner André:



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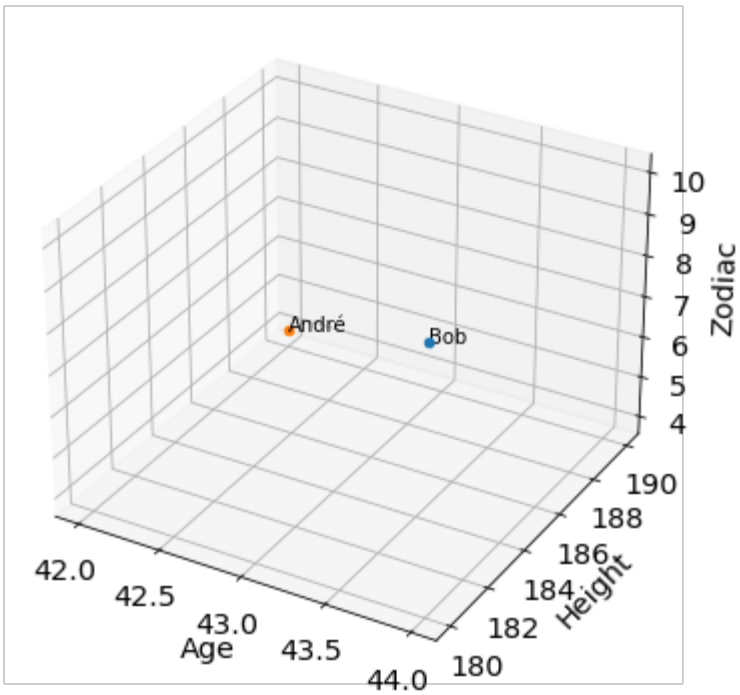
In [1]:
```
# see mpl_toolkits.mplot3d, in particular Axes3D
# add your code below
```



1.2*

Compute the Euclidean and Manhattan distances between the two points.

In [3]:
```
# add your code below
```

```
Euclidean distance is 11.832159566199232
Manhattan distance is 18
```

1.3*

Find the point that is half-way between the two points and plot it on the 3D plot you created in 1.1. What is the zodiac sign of the person who is described by that midway point?

In [4]:
```
# add your code below
```

```
[ 43. 185.   7.]
```

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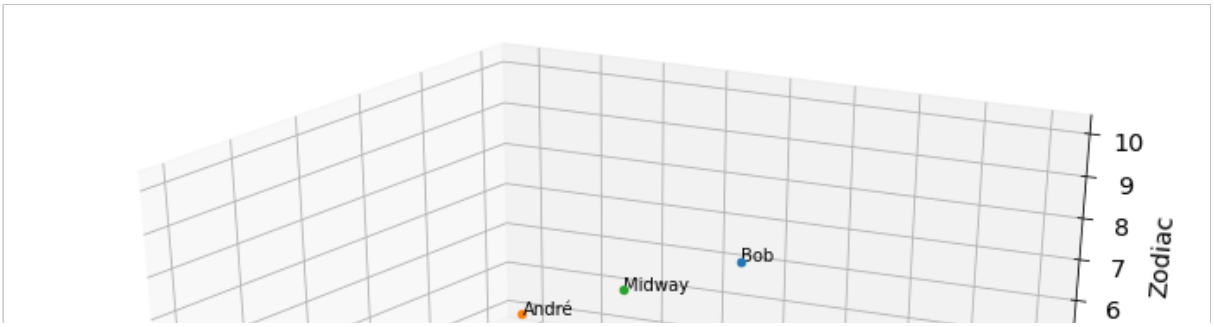
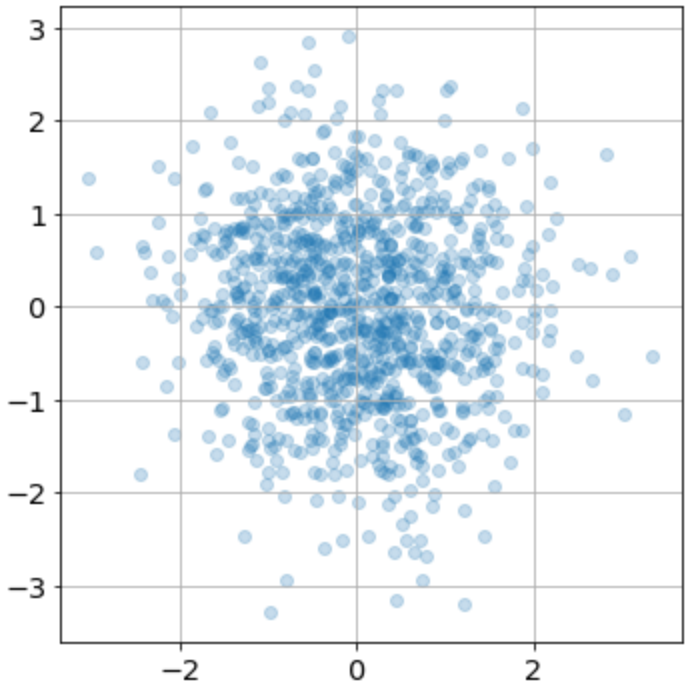


Answer: The zodiac sign is coded 7, which is *Gemini*!

1.4*

Now create 1000 two-dimensional vectors where each element is sampled from a normal distribution with mean 0 and standard deviation 1. Make a scatterplot of these points.

In [4]:
```
# see numpy.random.normal
# add your code below
```



1.5*

Multiply each one of these random vectors by the matrix [2 1; 1 1] and plot the resulting scatterplot.

In [5]:
```
# see numpy.dot
# add your code below
```

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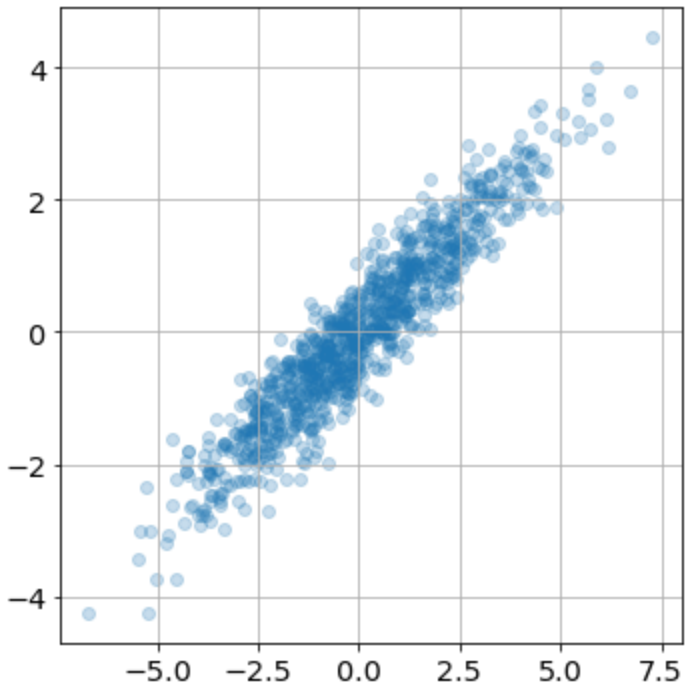

1.6*

Describe what has happened to the points plotted in #2.5.

Answer:

1.7*

For each of these points, multiply them now by the inverse of the matrix [2 1; 1 1] and make a scatter plot. It should look exactly the same as your plot in 2.4.

In [6]:

```
# see numpy.linalg.inv
# add your code below
```

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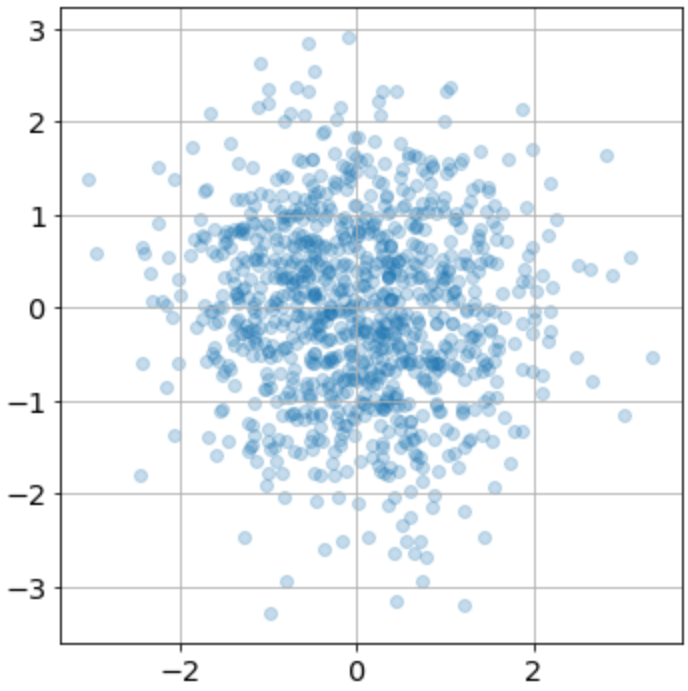

1.8*

Now create 1000, 2-dimensional points with each dimension uniformly distributed in [-1,1]. For each of these points, find the Euclidean distance to all other points. Make a histogram of the distribution of these distances. Set the limit of the xaxis to be between 0 and 10.

In [8]:

```
# see numpy.random.uniform
# add your code below
```

1.9

Now create 1000, 10-dimensional points in the same way as you created the points in #2.8. Now find the Euclidean distance to all other points in this collection. Make a histogram of the distribution of these distances.

In [11]:

```
# add your code below
```

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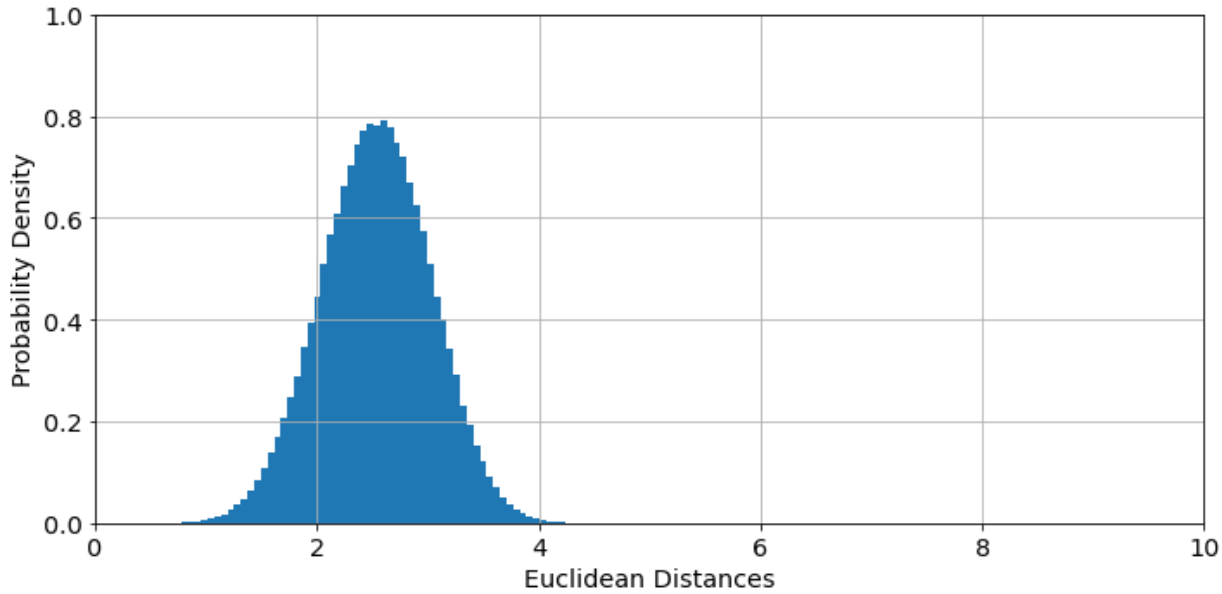
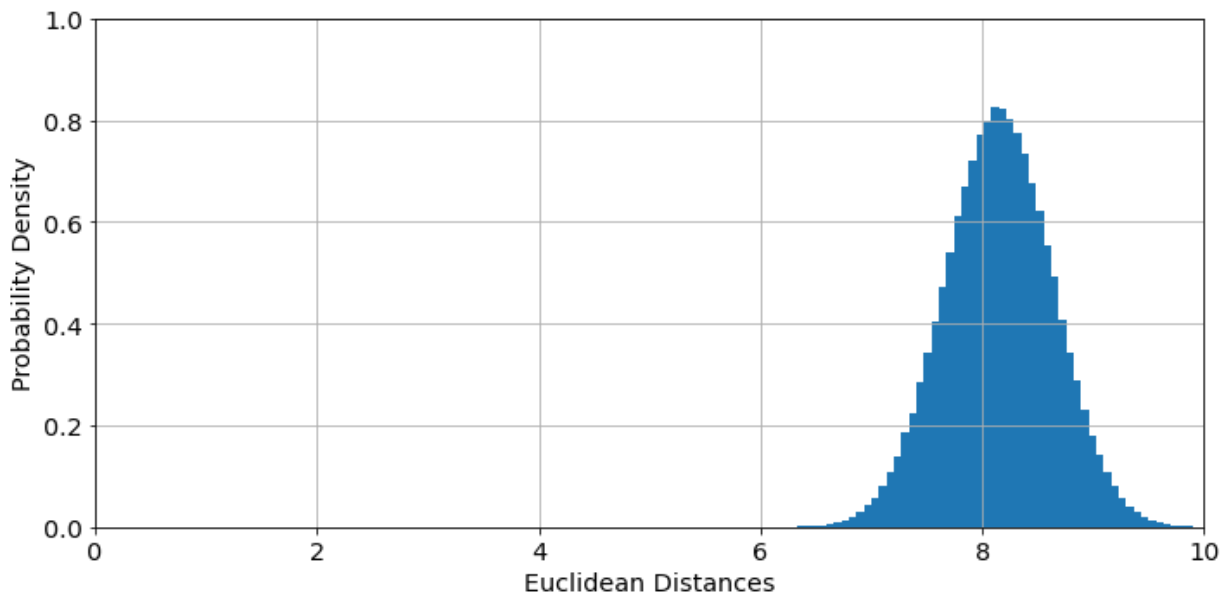


1.10

Now create 1000, 100-dimensional points in the same way as you created the points in #2.8. Find the Euclidean distance to all other points in this collection. Make a histogram of the distribution of these distances.

In [12]:
```
# add your code below
```



1.11

Describe what is happening to the distribution of distances on as you increase the number of dimensions.

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Answer:

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